Ashutosh Jarag

AI/ML Engineer | Data Engineer | Python · SQL · PySpark · LLMs · RAG · Databricks · Snowflake
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SUMMARY

AI & Data Engineer (PG-Diploma in Big Data Analytics, CDAC) with 14 months of internship experience. Built an **LLM pipeline that cut handwritten-bill data-entry effort by 70%**, a **RAG chatbot**, a **text-to-SQL assistant**, and end-to-end **medallion pipelines** on **PySpark, Snowflake, dbt** and **Databricks**. Works in **Python, SQL, LangChain, FastAPI, PostgreSQL** and **AWS**.

EXPERIENCE

AI/ML Intern Jan 2025 – Nov 2025
Compserv Consultant Pvt Ltd *Kolhapur, Maharashtra*

- Built an end-to-end **LLM pipeline** that reads handwritten bills, cleans and validates the extracted fields, lets users correct them in a review form and stores approved records in **PostgreSQL**, cutting manual data-entry effort by 70%.
- Created a **RAG** chatbot over the company website (**LangChain**, embeddings, vector database) and a **text-to-SQL** assistant that turns support-team questions into SQL queries, so staff can pull data without writing SQL.
- Developed a **secure file-storage and tracking application** with encryption and audit trails for company files and data, including its **FastAPI** backend and **PostgreSQL** storage, improving data traceability.
- Analyzed client datasets with **SQL** and **Python** (EDA, cleaning) to surface insights, and contributed to R&D that compared **LLMs, vector databases** and frameworks to choose the architecture for GenAI projects.

Data Science Intern Jun 2024 – Aug 2024
Tvarit Solutions Pvt Ltd *Pune, Maharashtra*

- Automated reporting with **Python pipelines** that pull data from **AWS** storage, **PostgreSQL** and flat files, preprocess it and generate business reports, replacing manual report preparation.
- Cleaned and explored **industrial data** (EDA, statistical analysis) to support senior engineers and data scientists in model and project development.
- Refactored procedural scripts into **modular Python scripts**, making the code easier to read, reuse and maintain.

TECHNICAL SKILLS

Programming Languages: Python, SQL (Joins, Window Functions), Bash/Shell Scripting
Generative AI & LLMs: LLMs, Generative AI, RAG, Prompt Engineering, LangChain, LangGraph, LangSmith, Gemini, Embeddings, Semantic Search, MCP, Vector Databases (Qdrant, ChromaDB, Pinecone, FAISS)
Machine Learning & NLP: Scikit-learn, XGBoost, PyTorch, NLP, Transformers, Hugging Face, Classification, Regression, Feature Engineering
Big Data & Data Engineering: Apache Spark, PySpark, Spark SQL, Databricks, Kafka, Hadoop, HDFS, Hive, ETL/ELT, Data Pipelines, Batch Processing, Parquet
Warehousing & Orchestration: Snowflake, Delta Lake, Star Schema, Dimensional / Data Modeling, SCD Type 2, Apache Airflow, dbt
Cloud, Backend & DevOps: AWS (S3), Azure (Fundamentals), GCP, FastAPI, REST APIs, Pydantic, Docker, Git, GitHub Actions (CI/CD), Linux
Databases & Analytics: PostgreSQL, MySQL, MongoDB, EDA, Tableau, Power BI

EDUCATION

Centre for Development of Advanced Computing (CDAC) Pune, Maharashtra
PG-Diploma in Big Data Analytics *Feb 2026 – Aug 2026*

Shivaji University Kolhapur, Maharashtra
B.Tech in Computer Science (Data Science) - 8.2 CGPA *Sep 2022 – May 2025*

Dr. D. Y. Patil Polytechnic Kolhapur, Maharashtra
Diploma in Computer Science - 84.23 % *Sep 2019 – May 2022*

PROJECTS

AI-Powered Research Paper Intelligence System | *Python, NLP, ML, LLM, RAG, ChromaDB, FastAPI* *GitHub*

- Built a research assistant over **3.1M+ arXiv papers**: trained a **TF-IDF + SVM** classifier to categorize papers, added **semantic search** and recommendations with **ChromaDB** embeddings, and used **Gemini** with **RAG** for summaries and document Q&A.
- Exposed everything through a modular **FastAPI** backend with **Pydantic**-validated endpoints and reusable service layers.

AI Math Agent | *Python, FastAPI, Gemini, Qdrant, RAG, MCP* *GitHub*

- Built a **RAG** math tutor on **Gemini** and **Qdrant** that retrieves knowledge-base context to generate step-by-step solutions; added **guardrails**, feedback-based retraining and an **MCP** web-search fallback.

Reliable Supply Chain Data Pipeline | *PySpark, Kafka, AWS S3, Snowflake, dbt, Docker, Tableau* *GitHub*

- Built a Bronze–Silver–Gold **medallion pipeline** on **S3 and PySpark** that deduplicates records, tracks history with **SCD Type 2**, routes bad records to a dead-letter store and writes partitioned **Parquet** with idempotent reruns.
- Modeled Gold tables with **dbt** on **Snowflake** External Tables and built **Tableau** dashboards for revenue, inventory, store and rider performance.

Big Data Sales Analytics Warehouse | *PySpark, SQL, Databricks, Delta Lake, Airflow, PostgreSQL* *GitHub*

- Modeled a **star schema** (fact and dimension tables) on **Databricks** and **Delta Lake**, wrote window-function **SQL** for KPIs, and orchestrated loads into **PostgreSQL** with **Airflow**.

CERTIFICATIONS & LEADERSHIP

Certifications: Cisco Data Analytics, Cisco Introduction to Data Science, Power BI, Advanced SQL
Achievements: Winner, AI/ML TechSphere college tournament; data team lead, Mahila Bachat Gat platform (KIC)